🌱 COLLEGE COMMUNITY ENGAGEMENT PROJECT (CEP)

EcoQuest — Campus Restoration Game

An Interactive Mobile Application for Fostering Student Sustainability Habits

Project Title	EcoQuest — Campus Restoration Game
Project Type	Community Engagement Project (CEP)
Project Theme	Gamified App for Sustainability
Technology Stack	React Native 0.86.3 • Expo SDK 57 • React 19.2.3
Target Platform	Android Mobile Application (Expo Go)
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Course Title & Code	**[Insert Official Course Title & Code — e.g., Community Engagement Project]**
Department	Department of Computer Science
Institution / College	**SMT. CHANDIBAI HIMATHMAL MANSUKHANI COLLEGE (CHM), Ulhasnagar-3**
Academic Year	2026–2027
Faculty Guide / Mentor	**[Insert Faculty Mentor Name & Designation]**
Date of Submission	**[Insert Submission Date]**

Executive Abstract

On university and college campuses, sustainability initiatives such as energy conservation, waste segregation, water stewardship, and low-carbon commuting are frequently communicated through passive notices, static signage, or administrative guidelines. While students recognize the importance of environmental conservation, the absence of immediate positive feedback, measurable individual progress, and visible collective results leads to low ongoing engagement.

EcoQuest — Campus Restoration Game is a mobile application developed as a Community Engagement Project (CEP) by **Aditya Deonarayan Jha**, Third Year B.Sc. Computer Science student at **SMT. Chandibai Himathmal Mansukhani College (CHM), Ulhasnagar-3**, to address this gap. Built using **React Native 0.86.3 and Expo SDK 57**, EcoQuest translates everyday campus eco-actions into an interactive **Campus Restoration Game**. Students complete micro-challenges—such as carrying a reusable water bottle, switching off classroom fans, taking sustainable transport, and segregating cafeteria waste—to earn verified **Lifetime Eco Points** and build a **Spendable Fund**. These points are reinvested into physical, sequential campus restoration projects (planting shade trees, installing rooftop solar panels, deploying smart compost hubs, and establishing pure water refill stations). As upgrades are funded, the central illustrated **Campus Restoration Scene** visibly evolves on-screen from a dormant quad into a thriving green sanctuary.

The application incorporates a two-tier points economy (210 total lifetime points available across 8 quests; 180 points required across 4 campus upgrades; 30 points final spendable surplus), 4 player levels, 8 achievement badges, a dynamic simulated leaderboard, offline local persistence via `AsyncStorage`, and a two-step demonstration reset feature. This report presents the background, objectives, system architecture, feature implementation, mathematical balance, automated test results, limitations, and future scope of the EcoQuest project.

1. Introduction

Environmental sustainability is among the most urgent global challenges, and higher educational institutions represent vital micro-communities for fostering sustainable habits. Everyday behaviors on campus—such as leaving fans running in unoccupied lecture halls, buying single-use bottled water, discarding unsegregated waste, and driving personal motorized vehicles for short inter-campus trips—collectively generate substantial carbon emissions, landfill waste, and resource depletion.

However, traditional campus sustainability campaigns rely heavily on top-down appeals, posters, or sporadic awareness drives. Behavioral science indicates that sustainable habits are adopted far more effectively when actions are:

- **1. Actionable & Achievable:** Broken down into small, daily micro-actions.
- **2. Immediately Rewarding:** Acknowledged with instant gamified feedback.
- **3. Visually Tangible:** Connected to a visible transformation in the student's immediate environment.

EcoQuest models this behavioral feedback loop. By framing campus sustainability as an interactive restoration project, students do not merely “read guidelines”—they actively rebuild a digital reflection of their campus quad through daily personal actions.

> 📌 **Note on Academic Demonstration Context**
>
> EcoQuest is an educational demonstration prototype developed for academic evaluation and community presentation. Its mechanics are designed to model incentive structures rather than measure physical environmental outcomes.

2. Problem Statement

Current university sustainability awareness efforts face three foundational obstacles:

- **1. Passive Information Delivery:** Guidelines posted on bulletin boards or university websites lack interactivity and fail to capture the attention of digital-native students.
- **2. The "Drop in the Ocean" Perception:** Individual students often feel that minor actions—such as turning off a switch or bringing a cloth bag—are inconsequential in the absence of visual feedback or aggregated recognition.
- **3. Absence of Tangible Progress:** Unlike academic coursework or sports, sustainability lacks milestone markers, progression levels, or public achievement indicators.

There is a distinct need for a lightweight, mobile-first demonstration tool that makes sustainability visible, structured, and rewarding within a student's daily routine without requiring expensive enterprise hardware or complex server infrastructure.

3. Project Objectives

The primary objectives of the EcoQuest project are:

1. **Promote Sustainability Awareness:** Introduce college students to actionable conservation habits across five core domains: Energy, Waste, Water, Transport, and Food.
2. **Structure Actions as Quests:** Present sustainability habits as clear, self-contained micro-tasks with explicit point values and impact descriptions.
3. **Implement a Two-Tier Gamified Economy:** Model an economy where *Lifetime Points* (which drive player level, badges, and leaderboard rank) are strictly distinguished from *Spendable Points* (which are spent to fund campus upgrades), ensuring that spending points never penalizes player standing.
4. **Visualize Environmental Restoration:** Create a dynamic, illustrated Campus Restoration Scene that visibly transitions through five cumulative stages (Stage 0 to Stage 4) as restoration projects are completed.
5. **Implement Multi-Tier Progression:** Incorporate four player levels, eight achievement badges with distinct criteria, and an eight-participant dynamic leaderboard.
6. **Ensure Reliable Local Persistence:** Enable game progress to automatically survive app reloads and restarts using local device storage (`AsyncStorage`).
7. **Provide Presentation Readiness:** Include a confirmed demonstration reset workflow that allows presenters to cleanly reset the app to baseline state during live demonstrations.

4. Scope of the Project

4.1 In-Scope Capabilities

- A standalone mobile application running on physical Android devices through Expo Go.
- Five functional navigation tabs: Home, Quests, Rank, Badges, and Profile.
- Eight predefined sustainability challenges across Energy, Waste, Water, Transport, and Food.
- Four sequential campus restoration upgrades costing 30, 40, 50, and 60 points respectively.
- A native illustration component rendered purely with React Native primitives and CSS layering.
- Local key-value storage using `@react-native-async-storage/async-storage`.
- Offline execution with zero external server, cloud backend, or login requirements.

4.2 Out-of-Scope (Explicit Boundaries)

- **No Automated Verification:** Quest completion is a simulated, self-reported user action; the app does not incorporate IoT sensors, camera AI, or GPS geofencing to physically verify real-world behavior.
- **No Live Multi-User Network:** The campus leaderboard features fictional benchmark peers to demonstrate real-time ranking algorithms without requiring live server connections.
- **No Cloud Account Synchronization:** User progress is stored locally on the demonstrator's device.
- **No Measured Scientific Impact:** Environmental impact labels are pedagogical estimates designed for demonstration rather than laboratory-certified carbon offsets.

5. System Requirements

5.1 Functional Requirements

ID	Requirement Description	Verification Method
FR-01	Provide five distinct tabs: Home, Quests, Rank, Badges, and Profile.	Inspected & Tested
FR-02	Display 8 challenges with point rewards, category icons, and category filter pills.	Inspected & Tested
FR-03	Tapping 'Mark as Done' awards points immediately, updates state, and blocks duplicate claims.	Automated Test (Node.js)
FR-04	Home screen displays Lifetime Points, Spendable Points, Quests Done, and Player Level.	Inspected & Tested
FR-05	Campus Restoration card displays next upgrade, cost, and affordability/shortfall feedback.	Automated Test (Node.js)
FR-06	Funding upgrade advances Campus Scene stage (0 to 4) and deducts spendable points.	Automated Test (Node.js)
FR-07	At Stage 4, purchase action is replaced with completion summary displaying 30-point surplus.	Automated Test (Node.js)
FR-08	Rank screen displays 8-participant leaderboard with deterministic tie-breaking and Top-3 podium.	Automated Test (Node.js)
FR-09	Badges screen evaluates 8 badges with distinct Earth Guardian (160) vs Level 4 (170) thresholds.	Automated Test (Node.js)
FR-10	Progress persists locally across app reloads under storage key `ecoquest.gameState.v1`.	Automated Test (Node.js)
FR-11	Profile screen provides two-step confirmation Reset Demo action restoring Stage 0.	Automated Test (Node.js)

5.2 Non-Functional & Environment Specifications

- **Zero Backend Dependency:** Fully operational offline without internet connectivity.
- **Responsive Mobile Layout:** Optimized for portrait Android phones with explicit safe-area clearance for Android 3-button navigation (48dp) and gesture navigation bars.
- **Zero External Graphic Dependencies:** Campus illustrations are rendered entirely via React Native primitives (`View`, `Text`) and CSS styling, ensuring zero image asset loading lag or broken asset links.
- **Atomic State Safety:** Sequential write queuing and synchronous ref tracking eliminate race conditions during rapid consecutive taps.

- **Development & Target Platform:** Windows 11 host with Android 10+ target via Expo Go, Node.js (v18+), React Native 0.86.3, and Expo SDK ~57.0.30.

6. Technology Stack

Component	Technology	Version	Role in EcoQuest Architecture
Framework	Expo SDK	~57.0.30	Managed workflow, rapid bundling via Metro
Mobile Runtime	React Native	0.86.3	Native Android layout primitives, View, Text, Alert
Core UI Library	React	19.2.3	Functional component lifecycle, useState, useEffect, useContext
Safe Area Insets	react-native-safe-area-context	~5.7.0	Handles Android status bar & 48dp bottom nav clearance
Local Storage	AsyncStorage	2.2.0	Offline key-value persistence under 'ecoquest.gameState.v1'
Status Bar	expo-status-bar	~57.0.1	Adaptive system bar typography styling

7. System Architecture

EcoQuest implements an **offline-first unidirectional data flow** managed by a centralized `GameProvider` context wrapper:

1. **App Launch & Hydration:** `GameProvider` reads `ecoquest.gameState.v1` from `AsyncStorage`. Raw data is validated via `validateSavedGameState()` to filter corrupt payloads and heal non-sequential upgrades. A themed loading card prevents UI flicker.
2. **State Mutations & Mutex:** Completing quests or purchasing upgrades updates synchronous refs and enqueues writes to a promise queue (`queuePersist`), ensuring atomic sequential disk writes.
3. **The Pure Selector Pattern:** Only two primary arrays are persisted: `completedQuestIds` and `purchasedUpgradeIds`. All downstream metrics—Lifetime Points, Spendable Fund, Stage, Level, Badges, and Ranks—are computed dynamically by pure selector functions, eliminating stale-state desynchronization.

8. Feature Implementation

- **Home Dashboard & Campus Scene:** Student profile banner (Aditya — Eco Explorer), animated level progress bar, impact metrics grid (Lifetime Points, Spendable Fund, Quests count), and the central 5-stage illustrated campus scene.
- **Sustainability Quests (`QuestsScreen.js`):** 8 categorized challenges across Energy (⚡), Waste (♻️), Water (💧), Transport (🚲), and Food (🥗) with one-tap completion and duplicate claim protection.
- **Campus Leaderboard (`RankScreen.js`):** Real-time standings ranking user among 7 benchmark peers (Maya, Aarav, Priya...), Top-3 synchronized podium, and deterministic tie-breaking.
- **Achievement Badges (`BadgesScreen.js`):** 8 trophy cards with dynamic progress counters, glowing unlocked badges, and distinct threshold separation.
- **Profile & Demo Reset (`ProfileScreen.js`):** Performance stats card and native `Alert.alert` confirmation dialog allowing instant, risk-free reset to Stage 0.

9. Game Rules and Mathematical Calculations

9.1 Quest Rewards Table (210 Lifetime Points Total)

ID	Title	Category	Points	Impact	Description
ch-1	Reusable Water Bottle	Waste ♻️	**30 pts**	Medium	Carry refillable bottle to classes today.
ch-2	Lights & Fans Off	Energy ⚡	**20 pts**	Low	Switch off switches in empty lecture halls.
ch-3	Green Commute	Transport 🚲	**40 pts**	High	Walk, cycle, or use campus shuttles.
ch-4	Plant-Based Meal	Food 🥗	**25 pts**	Medium	Choose vegetarian lunch at cafeteria.
ch-5	Segregate Campus Waste	Waste ♻️	**35 pts**	Medium	Separate wet organic and dry recyclables.
ch-6	5-Minute Shower	Water 💧	**25 pts**	Medium	Conserve hostel water during morning showers.
ch-7	Unplug Idle Chargers	Energy ⚡	**15 pts**	Low	Disconnect idle chargers to stop phantom drain.
ch-8	Bring a Tote Bag	Waste ♻️	**20 pts**	Low	Use a cloth tote bag for books & stationery.
TOTAL MAXIMUM ACHIEVABLE REWARDS			**210 pts**	**8 Sustainability Challenges**	

9.2 Campus Upgrades Table (180 Points Total)

Stage	Upgrade ID	Project Title	Cost	Visual & Physical Impact
Stage 1	`upg-trees`	Native Shade Trees	**30 pts**	Restores lush grass, shade canopy, and flora.
Stage 2	`upg-solar`	Rooftop Solar Array	**40 pts**	Adds photovoltaic panels on academic hall.
Stage 3	`upg-compost`	Smart Compost Hubs	**50 pts**	Deploys organic compost pods along paths.
Stage 4	`upg-water`	Pure Water Cistern	**60 pts**	Chilled water refill fountain in plaza.
TOTAL UPGRADE INVESTMENT REQUIRED			**180 pts**	**4 Cumulative Campus Upgrades**

> 💰 **Economic Surplus Invariant**
>
> Final Spendable Surplus = 210 Lifetime Points Earned − 180 Upgrade Points Spent = **30 Spendable Eco Points**.
> Lifetime Points never decrease upon purchasing upgrades, ensuring students are never demoted in level or leaderboard rank for funding environmental restoration.

9.3 Level & Badge Thresholds

- **Level 1: Eco Seedling:** 0 – 49 pts • **Level 2: Green Scout:** 50 – 99 pts • **Level 3: Earth Guardian:** 100 – 169 pts • **Level 4: Planet Champion:** 170 – 210 pts.
- **Earth Guardian Badge Distinction:** Unlocks at exactly **160 pts**, intentionally distinct from the Level 4 threshold of 170 pts.

10. User Interface and Workflow

The following figures mark the exact locations where device screenshots from the running Android phone in Expo Go should be inserted:

📷 [INSERT SCREENSHOT: Stage 0 Home Screen]

Capture directly from the Android phone running EcoQuest in Expo Go

- Shows dry, dormant campus quad illustration
- Level 1: Eco Seedling badge and 0 Eco Points displayed
- Native Shade Trees upgrade displayed as locked (Requires 30 pts)

Figure 1: Initial Baseline Dashboard & Dormant Campus Quad (Stage 0)

📷 [INSERT SCREENSHOT: Quests Screen]

Capture directly from the Android phone running EcoQuest in Expo Go

- Category filter pills (All, Energy, Waste, Water, Transport, Food)
- Challenge cards with point badges and 'Mark as Done' buttons
- Demonstration disclaimer banner at top

Figure 2: Sustainability Quests Catalog with Category Filters

📷 [INSERT SCREENSHOT: Upgraded Campus Scene (Stage 1 / Stage 4)]

Capture directly from the Android phone running EcoQuest in Expo Go

- Campus Scene visibly displays restored trees and rooftop solar panels
- Restoration Card shows next project or Stage 4 completion banner
- Spendable points deducted while lifetime points remain at high-water mark

Figure 3: Campus Restoration in Progress (Restored Green Infrastructure)

📷 [INSERT SCREENSHOT: Rank Screen & Podium]

Capture directly from the Android phone running EcoQuest in Expo Go

- User climbing dynamically from Rank #8 toward Rank #4
- Synchronized Gold, Silver, and Bronze podium blocks
- Deterministic tie-breaking demonstrated with benchmark peers

Figure 4: Campus Leaderboard & Top-3 Synchronized Podium

📷 **[INSERT SCREENSHOT: Badges Showcase Grid]**

Capture directly from the Android phone running EcoQuest in Expo Go

- Two-column grid displaying unlocked and locked trophies
- Live progress counters on domain mastery badges (e.g., '1 / 2 Waste quests')
- Distinct status for Earth Guardian (160 pts) and Planet Hero (200 pts)

Figure 5: Achievement Badges Showcase Grid

📷 **[INSERT SCREENSHOT: Profile Screen & Reset Dialog]**

Capture directly from the Android phone running EcoQuest in Expo Go

- Demonstrator profile card and cumulative performance statistics
- Native Alert.alert confirmation modal popup
- 'Cancel' and 'Reset Everything' demonstration buttons clearly visible

Figure 6: Student Profile & Two-Step Demo Reset Dialog

11. Testing and Verification Results

> 🧪 **Automated Master Integration Test Suite (tests/test_integration.mjs)**
> • Total Assertions Executed: **83**
> • Assertions Passed: **83**
> • Assertions Failed: **0**
> • Success Rate: **100% PERFECT PASS RATE**
> • Areas Validated: Invariants (210/180/30), 4 Levels, 8 Badges, Leaderboard Tie-Breaks (#8 to #4), Storage Schema Recovery (Null/Corrupt/Non-Sequential), and Full Demo Reset.

Metro Android Compilation: Bundled cleanly at `http://localhost:8081/index.bundle?platform=android&dev=true` with **HTTP 200 OK** across 763 modules with zero syntax warnings.

Physical Android Device Verifications: Hydration loading card verified, real-time button transitions confirmed, cumulative artwork transitions validated across all 5 stages, and safe-area clearance verified for 48dp bottom navigation pills.

12. Limitations

- **1. Self-Reported Actions:** Quest completions are self-reported by the user tapping in-app buttons; there is no physical verification via camera AI, barcode scanning, or IoT sensor integration.
- **2. Simulated Benchmark Leaderboard:** The seven competitor profiles on the leaderboard are fictional benchmark models designed to demonstrate real-time ranking algorithms without requiring multi-user database infrastructure.
- **3. Offline Client Architecture:** Storage is strictly local to the device using `AsyncStorage`. There is no cloud backend, relational database, user authentication system, or cross-device account sync.
- **4. Illustrative Environmental Impact:** Stated environmental impacts are educational estimates designed to contextualize sustainable habits, not certified scientific carbon calculations.

13. Future Scope

- **1. Centralized Cloud Backend:** Introducing a Node.js / PostgreSQL backend with user authentication to enable cross-device synchronization and genuine campus-wide student participation.
- **2. Multiplayer Live Leaderboards:** Real-time inter-hostel or inter-department sustainability competitions.
- **3. IoT & QR Verification:** Integrating QR codes at campus waste bins, water refill stations, and bicycle racks to physically verify student sustainability actions.
- **4. Campus Administration Dashboard:** A web dashboard for university sustainability committees to aggregate anonymous data on campus resource conservation trends.

14. Conclusion

The **EcoQuest — Campus Restoration Game** successfully demonstrates how gamification principles can transform passive campus sustainability guidelines into an engaging, interactive mobile experience.

By combining an intuitive two-tier points economy, five cumulative campus restoration stages, multi-tier progression, dynamic leaderboard competition, and robust local persistence, the application provides an educational and reliable presentation tool. Developed with React Native and Expo SDK 57 by **Aditya Deonarayan Jha** at **SMT. CHM College, Ulhasnagar-3**, EcoQuest achieves high visual appeal, zero broken asset dependencies, and complete presentation readiness for the College Community Engagement Project evaluation.

References

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